DATA ANALYTICS FOUNDATION FOR MACHINE LEARNING

ASSESSMENT QUESTION PAPER

Statistics • Probability • NumPy • Pandas • EDA

Duration	MaximumMarks	AssessmentType
2Hours 30Minutes	80	Written/Analytical

INSTRUCTIONS

Answer all questions.
Show calculations wherever applicable.
Support conclusions with reasoning or calculated values.
Python/Pandas syntax may be used where required; marks emphasize logic and understanding.
Do not assume that correlation implies causation.
Write your answers in the separate answer sheets provided.

SECTION A – CONCEPTUAL UNDERSTANDING $10 \times 2 = 20$

Q1. Mean vs Median – Outlier Effect

A company's employee salaries are:
₹25,000 | ₹28,000 | ₹30,000 | ₹32,000 | ₹35,000 | ₹2,00,000
Which measure better represents the typical employee salary: mean or median? Give one reason. [2]

Q2. Standard Deviation

ClassA				
68	69	70	71	72.
ClassB				
40	55	70	85	100

Both classes have the same mean. What does standard deviation tell you about these two classes? [2]

Q3. Probability vs Conditional Probability

Explain the difference between P(Placed) and P(Placed | Internship) using a student-placement example. [2]

Q4. False Positive

A fraud detection system flags a genuine transaction as fraudulent. What type of error is this? Why can many false positives be a problem for a business? [2]

Q5. NumPy vs Pandas

State one important difference between a NumPy array and a Pandas DataFrame. Give one situation where Pandas is more suitable. [2]

Q6. Missing Values

A dataset contains 10,000 student records. The Attendance column has 150 missing values. Should you automatically replace all missing values with zero? Explain. [2]

Q7. Duplicate Records

Student_ID	Name	Department
101	Arun	CSE
102	Bala	ECE
101	Arun	CSE

A student dataset contains the records below. Is the repeated Student_ID necessarily a data error? What should you check before removing it? [2]

Q8. Correlation

Attendance and marks have correlation = 0.82. Can the analyst conclude that increasing attendance will definitely increase marks? Explain. [2]

Q9. EDA

Why should EDA be performed before building an ML model? Mention any two reasons. [2]

Q10. ML Readiness

A dataset has missing values, duplicate records, inconsistent data types and extreme values. A student immediately trains an ML model. Is this good practice? Give two reasons. [2]

SECTION B – APPLIED ANALYTICAL THINKING $6 \times 5 = 30$

Q11. Statistical Comparison [5 Marks]

Calculate the following for both subjects. Then decide which subject shows greater variation and support your answer.

Subject	Marks
Python	60|65|70|75|80|100
SQL	68|69|70|71|72|70

Measure	Python	SQL
Mean	600	70
Median	70	70
Minimum	65	
Maximum		

Calculate mean, median, minimum and maximum for both. [3]

Which subject shows greater variation? Support using your values. [2]

Q12. Outlier Investigation [5 Marks]

A placement team records the following aptitude scores:

Student	1	2	3	4	5	6	7	8	9
AptitudeScore	45	48	50	52	49	51	47	50	95

Calculate mean and median. [2] b) Which value appears unusual? [1] c) Explain why mean and median behave differently. [2]

Q13. Probability – Placement Analysis [5 Marks]

A college has 100 students. Use the table below.

Placement/Internship	Internship	NoInternship	Total
Placed	40	10	50
NotPlaced	30	20	50
Total	70	30	100

Calculate: a) P(Placed) [1] b) P(Placed | Internship) [1] c) P(Internship | Placed) [1] d) Are the two conditional probabilities the same? Explain. [2]

Q14. Pandas – Student Dataset [5 Marks]

Use the following DataFrame and write Pandas statements/logic to answer (a)–(e).

Student_ID	Department	Attendance	Python	SQL
101	CSE	82	85	78
102	ECE	68	72	80
103	CSE	91	88	92
104	EEE	75	65	70
105	CSE	89	95	90

Attendance >75%. b) Python >80. c) Department-wise average Python. d) Sort by Python descending. e) Student with highest SQL. [1 mark each]

Q15. Data Cleaning [5 Marks]

The placement department provides the following student data. Identify at least four data-quality problems and explain how you would handle each.

Student_ID	Attendance	Python	Department
101	85	78	CSE
102	92	85	CSE
103	110	75	ECE

Student_ID	Attendance	Python	Department
104	NA	82	EEE
102	92	85	CSE
105	78	105	CSE

Consider: range validation, missing values, duplicate records, data types and invalid marks. [5]

Q16. Choosing the Correct Visualization [5 Marks]

For each requirement, name a suitable visualization and briefly justify your choice.

Requirement	YourVisualization	Why?
a)Numberofstudentsineachdepartment	bar chart	
b)RelationshipbetweenCGPAandplacement	~~scatter~~	
c)Distributionoftechnicalscores		
d)Placementrate:InternshipvsNoInternship	Scatter Plot	
e)Relationshipbetweenattendanceandtechnicalscore		

[1 mark each]

SECTION C – INTERVIEW-STYLE CASE ANALYSIS 3 × 10 = 30

These questions intentionally provide less step-by-step guidance. Focus on reasoning,interpretation and defensible conclusions.

Q17. Placement Dataset – Investigate Before Concluding [10 Marks]

A placement team says: "Students with internships have a much higher placement rate. Therefore, internship is the main reason students get placed."

Group	Students	Placed	AverageCGPA	AverageTechnicalScore
Internship	200	150	7.8	78
NoInternship	300	120	7.1	65

Calculate placement percentage for both groups. [2]

Does the data support a higher placement rate for internship students? [2]

Can we conclude internship causes higher placement? Explain. [2]

Identify two other variables that could influence placement. [2]

What additional analysis would you perform before a final recommendation? [2]

Q18. EDA Case Study – Find the Story [10 Marks]

You receive 5,000 student records with the following fields:

Student_ID	Department	Gender	CGPA	Attendance	Study_Hours
…	…	…	…	…	…

Internship	Projects	Backlogs	Aptitude_Score	Technical_Score	Communication_Score	Placed
…	…	…	…	…	…	…

The manager asks whether historical data can be used to understand factors associated with placement and eventually build a prediction model. Do not build the ML model yet.
What would you inspect first? [2] b) Give at least three data-quality checks. [2] c) Give three relationships to investigate. [2] d) Which visualizations would you use? [2] e) What is the likely target and is the future problem classification or regression? Explain. [2]

Q19. Real Interview-Style Analytical Challenge [10 Marks]

A company uses an AI system to identify fraudulent transactions. Use the following information:

Construct the confusion matrix. [4]

	ActuallyFraud	ActuallyGenuine
FlaggedFraud		
NotFlagged		

Calculate Precision. [2]

c) Calculate False-Positive Rate. [2]

d) Management says: "The system identifies 90% of fraudulent transactions, so it is highly reliable." Do you agree? Explain using your statistics. [2]

TOTAL: 80 MARKS

Assessment intent: Evaluate conceptual understanding, calculation accuracy, data-manipulation logic, data-quality awareness, EDA reasoning and independent analytical thinking before students enter Machine Learning.